

# Global Comparison Report

Monte Carlo Strategy Paired-Difference Analytics

|                                 |                      |                                  |
|---------------------------------|----------------------|----------------------------------|
| GENERATED                       | MONTE CARLO RUNS     | STRATEGIES TESTED                |
| 10 May 2026 13:40               | 1,000                | 4                                |
| METHOD                          | UNIT                 | REPORT TYPE                      |
| Discrete-Event Simulation (DES) | Haemodialysis Centre | Comparative Performance Analysis |

## Abstract

*This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 1,000 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.*

**Keywords:** Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

## 1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

### TEMPORAL BOUNDS

| Parameter        | Value | Unit       | Description                                     |
|------------------|-------|------------|-------------------------------------------------|
| Shift Duration   | 300   | <i>min</i> | Fixed -- hard boundary for each simulated shift |
| Machine Cooldown | 60    | <i>min</i> | Mandatory inter-session machine rest period     |

### RESOURCE RANGES (uniform draw per iteration)

| Parameter           | Value     | Unit            | Description                          |
|---------------------|-----------|-----------------|--------------------------------------|
| Total Machines      | 15 - 20   | <i>units</i>    | Physical dialysis stations available |
| Patient Volume      | 15 - 20   | <i>patients</i> | Demand sampled each shift            |
| Nursing Staff       | 2 - 4     | <i>nurses</i>   | Concurrent supervision capacity      |
| Session Duration    | 240 - 360 | <i>min</i>      | Per-patient dialysis time            |
| Machine Ready Delay | 0 - 90    | <i>min</i>      | Post-cooldown preparation lag        |

### STOCHASTIC EVENT GENERATORS

| Parameter              | Value           | Unit       | Description                            |
|------------------------|-----------------|------------|----------------------------------------|
| Patient Arrival Offset | Uniform[0, 60]  | <i>min</i> | Arrival jitter relative to shift start |
| Setup Duration         | Uniform[10, 20] | <i>min</i> | Machine preparation time per patient   |

### PROBABILITY METRICS

| Parameter                  | Value | Unit | Description                                        |
|----------------------------|-------|------|----------------------------------------------------|
| Machine Defect Probability | 15.0% | --   | Independent Bernoulli draw per machine per session |

### EXPERIMENT PARAMETERS

| Parameter              | Value | Unit        | Description                           |
|------------------------|-------|-------------|---------------------------------------|
| Monte Carlo Iterations | 1,000 | <i>runs</i> | Independent replications per strategy |